Bridge Day 4 STUDENT QUIZ

Comparisons, Exact Boundaries, and Ordered Branches

Learning sequence: Predict - Trace - Explain - Test - Interpret

Thursday, July 9, 2026 • 20 points • target 16 minutes

Name		**UIN**		**Section**	

Assessment boundary and evidence source

Contract: 07a04826c975

Cumulative foundations: input conversion, comparisons, inclusive boundaries, if/elif/else, f-strings

Scaffold level: complete ordered four-case classifier

Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.13, 18.15, 18.16, 18.17.

Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Evaluate exact-boundary comparisons. - 5 points

```
low = 10
high = 15
value = 15
within = low <= value <= high
above = value > high
print(within, above)
```

Question: Evaluate every comparison and explain which operator controls the upper boundary.

Item 2. Trace the first matching ordered branch. - 5 points

```
reading = 25

if reading > 25:
    status = "high"
elif reading >= 20:
    status = "normal"
else:
    status = "low"

print(status)
```

Question: Name the first matching branch, give exact output, and state what happens at 25 and 25.1.

Complete program - 10 points

- Read temperature as a float.
- Print invalid when temperature is below -40 or above 80.
- For valid input, classify cold below 15, warm from 15 through 30 inclusive, and hot above 30.
- Print exactly Status: followed by the classification using one if/elif/elif/else chain.

Program workspace

Test input, expected result, and why this test matters